Houlin (Richard) Wang

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Computational linguist and speech scientist with 5+ years of experience in ASR evaluation, forced alignment, and corpus analysis for low-resource and non-standard speech

Education

Doctor of Philosophy (Ph.D.) in Linguistics — Expected June 2027
University of California, Santa Cruz — Santa Cruz, CA
Specialization in Speech Technology & Computation

Master of Arts (M.A.) in Linguistics, GPA 3.97 — June 2022
University of California, Los Angeles — Los Angeles, CA

Bachelor of Arts (B.A.) in Linguistics, GPA 3.95, Magna Cum Laude — June 2022
Specialization in Computation • Japanese minor • Department Scholar and Departmental Honors

Experience

Computational Linguistics Researcher — September 2022 – Present
University of California, Santa Cruz — Santa Cruz, CA

- Build Python/R pipelines for acoustic feature extraction, annotation, and quality control at 10x manual scale
- Run mixed-effects regression and ML classification across 20,000+ model fits and 8 variables, showing duration does not predict consonant lenition in Mandarin
- Develop a lenition classifier on public Mandarin corpora (AISHELL-3, THCHS-30) for corpus quality monitoring
- Design online speech-perception experiments (80+ participants) and model listener decision boundaries in R

AI Evaluation Researcher — June 2026 – Present
Living Tongues Institute for Endangered Languages — Remote

- Evaluate Meta's Omnilingual ASR on 5 low-resource Munda languages (300M–7B models), scoring WER, CER, and phone matching against human IPA
- Model 1,557 aligned Ho phone pairs with Random Forest/XGBoost (AUC > 0.98): errors track vowel quality
- Measure Ho phone error: 50.7% (strict) vs 39.9% (broad), separating notation mismatch from true errors
- Diagnose decoder looping, wrong-script output, and mislabeled training data; annotate 90+ hours of audio

Research Mentor & Instructor — June 2026 – August 2026
Science Internship Program, University of California, Santa Cruz — Santa Cruz, CA

- Designed and taught an 8-week curriculum in Python, R, Praat, and hypothesis testing (80 hours total)
- Mentored 3 student researchers (10 hours/week) in annotating and modeling 2,500+ speech tokens

Computational Linguistics Research Assistant — September 2019 – June 2021
UCLA Computational Phonology & Language Processing Labs — Los Angeles, CA

- Built Python algorithms to analyze phonotactic patterns across 10,000+ lexicon entries
- Created reproducible R pipelines for corpus analysis and coordinated protocols for 30+ study participants

Projects

Mandarin Word-List Aligner: forced alignment and lenition classifier (Python, MFA, wav2vec 2.0)
ASR Error Metrics: WER/CER with alignment and substitution analysis (Python, unit-tested)

Skills

Speech & NLP: ASR evaluation (WER, CER), forced alignment (MFA), Praat, corpus annotation
ML & Statistics: Mixed-effects regression (lme4), ML classification, bootstrapping, hypothesis testing
Programming: Python (pandas, NumPy, SciPy), R (tidyverse), SQL, PyTorch, Hugging Face, Git, Bash, LaTeX
Languages: Mandarin (native), English (fluent), German and Japanese (working)

Publications

Wang, R. (2025). Distribution of tone affects prosodic parsing in Beijing Mandarin. *Proceedings of AMP, 1*(1)
Wang, R. (2025). Morphosyntax-prosody mismatch in Beijing Mandarin. *Proceedings of WCCFL 41*, 592–599
Wang, R. et al. (under review). Does Omnilingual work? Testing ASR on Munda languages. *ComputEL-10*

Awards

THI Summer Fellowship ($6,000) • THI Dissertation Fellowship ($5,000) • SIP Fellowship ($4,750), all 2026